

Financial Research AI Agent Development Project – Dual Track Version

Team Size: 3–4 students | **Duration:** 8 Weeks | **Choose Your Challenge Level**

Project Overview

Build an AI-powered financial research assistant that analyzes stock markets, economic trends, and provides personalized investment insights through intelligent data integration. You'll master financial APIs, sentiment analysis, and risk assessment while creating a production-ready fintech application.

Domain Focus: Financial Markets & Investment Analysis
Core Skills: Financial data processing, sentiment analysis, risk modeling, regulatory compliance

CHOOSE YOUR TRACK

Track A: Essential (Recommended for Most Teams)

Focus: Core financial analysis skills, achievable goals, strong fundamentals

- **Domain:** Stock Analysis OR Personal Finance Helper
- **APIs:** 2–3 basic integrations (Yahoo Finance + News API + Economic data)
- **Framework:** Standard LangChain with financial tools
- **UI:** Streamlit with financial charts
- **Deployment:** Single platform (Streamlit Cloud)
- **Data:** Basic stock analysis and news sentiment

Track B: Advanced (For Confident Teams)

Focus: Industry-ready fintech skills, complex financial modeling

- **Domain:** Comprehensive Financial Research Platform
- **APIs:** 5+ integrations with fallback strategies and rate limiting
- **Framework:** LangGraph for complex financial workflows
- **UI:** Professional React/Next.js with advanced charting

- **Deployment:** Multi-service architecture with real-time data
- **Features:** Portfolio optimization, risk modeling, regulatory compliance

Learning Outcomes

By completing this project, you will:

- **Financial Data Mastery:** Connect with global and Indian market APIs for real-time analysis
- **AI-Powered Insights:** Implement sentiment analysis and predictive modeling for investment decisions
- **Risk Assessment:** Build sophisticated algorithms for portfolio risk evaluation
- **Regulatory Awareness:** Handle financial data with proper security and compliance
- **Real-time Processing:** Create systems that process live market data and news
- **Professional Fintech Skills:** Deploy production-ready financial applications

Indian Financial Market APIs & Data Sources

Primary Indian Market APIs:

- **NSE India:** National Stock Exchange official API
- **BSE API:** Bombay Stock Exchange data
- **Zerodha Kite API:** Popular trading platform API
- **Alpha Vantage:** Supports Indian stocks (NSE/BSE)
- **Financial Modeling Prep:** Indian stock fundamentals
- **RBI API:** Reserve Bank of India economic indicators

News & Sentiment (India-focused):

- **Economic Times API:** Leading financial news
- **Moneycontrol:** Indian market news and analysis
- **BloombergQuint:** Business news API
- **NewsAPI:** Filter for Indian financial news

Economic Data:

- **RBI Database:** Interest rates, inflation, monetary policy
- **Ministry of Statistics:** GDP, industrial production
- **SEBI Data:** Market regulations and filing

Week-by-Week Structure

Week 1-2: Foundation & Quick Win

Goal: Get basic financial data flowing and deployed early

Track A Checklist:

- ☐ Create GitHub repo with financial project structure
- ☐ Set up development environment (Python, pandas, yfinance, LangChain)
- ☐ Build basic stock price chatbot with Indian stock support (RELIANCE.NS, TCS.NS)
- ☐ Add simple chart visualization using Plotly/Streamlit
- ☐ Deploy basic version on Streamlit/Gradio Cloud
- ☐ Record 2-minute demo showing stock price queries
- ☐ **Milestone:** Working deployed demo with Indian stock data

Track B Checklist:

- ☐ All Track A requirements plus:
- ☐ Set up advanced environment (LangGraph, Redis for caching, PostgreSQL)
- ☐ Implement real-time data streaming architecture
- ☐ Add comprehensive error handling for API failures
- ☐ Set up monitoring for API rate limits and costs
- ☐ Create CI/CD pipeline with financial data testing

Week 3-4: Core Financial Agent Architecture

Goal: Transform into a multi-source financial analysis agent

Track A Checklist:

- ☐ Integrate 2 financial tools (stock data + news sentiment)
- ☐ Implement basic financial calculations (moving averages, RSI)
- ☐ Add simple news sentiment analysis using TextBlob/VADER
- ☐ Create stock comparison functionality
- ☐ Handle API rate limiting and caching
- ☐ **Milestone:** Agent provides multi-dimensional stock analysis

Track B Checklist:

- ☐ All Track A requirements plus:

- ☐ Implement complex agent workflow with LangGraph for financial research
- ☐ Add 4+ financial tools (stocks, options, futures, bonds, economic indicators)
- ☐ Implement sophisticated sentiment analysis using transformer models
- ☐ Add portfolio tracking and performance analytics
- ☐ Create comprehensive financial data pipeline with ETL processes
- ☐ Implement advanced technical analysis indicators

Week 5-6: Domain Specialization

Track A Options (Choose ONE):

Option A1: Indian Stock Analysis Assistant

- ☐ Integrate NSE/BSE data via Alpha Vantage or Yahoo Finance (.NS/.BO suffixes)
- ☐ Add SQLite database for saving stock watchlists
- ☐ Create basic fundamental analysis (P/E, debt ratios, growth metrics)
- ☐ Implement sector comparison for Indian markets
- ☐ Add INR currency handling and Indian market hours

Option A2: Personal Finance Helper

- ☐ Create expense tracking with categories
- ☐ Add SIP (Systematic Investment Plan) calculator
- ☐ Implement basic goal-based investment suggestions
- ☐ Add Indian tax implications (LTCG/STCG calculations)
- ☐ Create simple portfolio tracker

Track B Requirements:

- ☐ Integrate 5+ financial APIs with sophisticated error handling
- ☐ Design complex database schema for financial data (PostgreSQL)
- ☐ Implement advanced portfolio optimization using Modern Portfolio Theory
- ☐ Add real-time market alerts and notification system
- ☐ Create multi-asset class analysis (equities, fixed income, commodities)
- ☐ **Advanced:** Implement options pricing models or algorithmic trading signals

Week 7-8: Polish & Production

Track A Checklist:

- ☐ Create professional Streamlit interface with:
 - ☐ Interactive stock charts with technical indicators
 - ☐ News sentiment dashboard
 - ☐ Portfolio tracking cards
 - ☐ Export functionality (PDF reports)
- ☐ Add input validation for stock symbols and date ranges
- ☐ Implement proper error messages for market closures
- ☐ Write comprehensive README with Indian stock symbol examples
- ☐ Prepare final demo video (5-7 minutes) showcasing Indian stock analysis
- ☐ **Milestone:** Production-ready financial analysis application

Track B Checklist:

- ☐ Build professional React/Next.js interface with advanced charting (Chart.js/D3)
- ☐ Deploy with proper microservices architecture (API gateway, data services)
- ☐ Add comprehensive monitoring dashboard for financial metrics
- ☐ Implement financial data security best practices (encryption, secure APIs)
- ☐ Add performance optimization for real-time data processing
- ☐ Create technical documentation including API documentation
- ☐ Prepare comprehensive demo (8-10 minutes) with live trading scenarios

Assessment Criteria

Track A Grading (100 points):

- **Financial Functionality (40 points):** Working agent with 2+ financial tools + database
- **Data Accuracy (20 points):** Correct financial calculations and Indian market handling
- **User Interface (15 points):** Clean, functional Streamlit app with charts
- **Deployment (15 points):** Stable hosted application with live data
- **Documentation (10 points):** Clear README with financial setup instructions

Track B Grading (120 points + bonuses):

- **Advanced Financial Architecture (45 points):** Complex workflows, portfolio optimization

- **Professional FinTech UI/UX (25 points):** Advanced charting, real-time updates
- **Production Deployment (20 points):** Multi-service architecture with monitoring
- **Financial Code Quality (15 points):** Testing, security, regulatory considerations
- **Technical Presentation (15 points):** Financial modeling explanation, risk analysis
- **Bonus Features (up to 20 points):** Options pricing, algorithmic signals, mobile app

Suggested Tech Stacks

Track A Stack:

- **Frontend:** Streamlit with Plotly for financial charts
- **Backend:** Python + LangChain + pandas + yfinance
- **Database:** SQLite for portfolio and watchlist storage
- **LLM:** Gemini (free tier) or GPT-3.5-turbo
- **Deployment:** Streamlit Cloud
- **Financial APIs:** Yahoo Finance, Alpha Vantage (free), NewsAPI

Track B Stack:

- **Frontend:** React/Next.js + Tailwind CSS + Chart.js/D3
- **Backend:** FastAPI + LangGraph + pandas + asyncio for real-time data
- **Database:** PostgreSQL + Redis (market data caching)
- **LLM:** GPT-4 or Claude for sophisticated financial analysis
- **Deployment:** Vercel (frontend) + Railway/Render (backend)
- **Monitoring:** Custom financial dashboard with LangSmith
- **Financial APIs:** Multiple with failover strategies

Success Metrics

Track A Success:

- 90%+ teams deploy working financial application with Indian market support
- 80%+ teams demonstrate accurate financial calculations and sentiment analysis
- All teams can explain basic financial concepts and API integrations

- Portfolio-worthy fintech project for internship applications

Track B Success:

- Production-grade fintech application with real-time capabilities
- Industry-standard financial data security and compliance practices
- Advanced financial modeling and risk assessment implementation
- Interview-ready technical presentation for fintech roles

Submission Requirements

All Teams Submit:

GitHub Repository with:

- Complete source code with financial data processing
- README with setup instructions and API key configuration
- Demo video (5-10 minutes) showcasing financial analysis capabilities
- Architecture documentation explaining financial data flow
- Live deployment with working URL and sample portfolios

Viva Presentation covering:

- Financial API integration strategies and challenges
- Technical architecture decisions for handling market data
- Risk assessment and compliance considerations
- Individual team member contributions and learning outcomes
- Future improvement ideas for enhanced financial features

Track B Additional Requirements:

- Technical blog post about fintech development challenges
- Performance benchmarks for real-time data processing
- Security assessment documentation for financial data handling
- API documentation for your financial endpoints

Portfolio Benefits:

- **LinkedIn Showcase:** Help teams create professional fintech project posts
- **GitHub Stars:** Instructor promotion of best financial repositories
- **Industry Connections:** Introduction to Indian fintech companies and startups

- **Internship Opportunities:** Showcase to companies like Zerodha, Groww, Paytm Money

Frequently Asked Questions

Q: How do I handle Indian stock symbols and market timings?

A: Use .NS suffix for NSE stocks (RELIANCE.NS) and .BO for BSE (RELIANCE.BO). Account for Indian market hours (9:15 AM – 3:30 PM IST).

Q: What about regulatory compliance for financial advice?

A: Your agent should provide analysis, not investment advice. Always include disclaimers and encourage users to consult financial advisors.

Q: How do I manage API costs for real-time financial data?

A: Track A uses free tiers (Yahoo Finance, Alpha Vantage free plan). Track B teams should implement caching and rate limiting strategies.

Q: Can we switch tracks mid-project?

A: Yes, from B to A after Week 4 checkpoint. Consider complexity of financial modeling when deciding.

Q: How do I handle market holidays and weekends?

A: Implement proper error handling for non-trading days and provide cached data or historical analysis during closures.

Q: What about cryptocurrency integration?

A: Advanced teams can add crypto APIs, but focus on traditional markets first for core requirements.